Homework 6

Contents

Exercise 1 Conjugate Gradient — Feature 1

Implement the Algorithm 5.2 (conjugated gradient) in the book Numerical Optimization of Nocedal and Wright to solve the system

$$Ax = b \quad (1)$$

where $A \in \mathbb{R}^{n \times n}$ such that $A^\top = A$ and $A \succ 0$, $b \in \mathbb{R}^n$

Apply the system to solve a system of equations: such as
- A given in the file `M_sys125x125.txt`,
- b given in the file `V_sys125x1.txt`,

using a preconditioner different from the Identity.

We use the Jacobi preconditioner, which consists in taking the diagonal elements of A, and forming a diagonal matrix $\hat{A}$ with these elements. This matrix is easy to invert, so solving $\hat{A}Ax = \hat{A}b$ is cheap.

To execute the program, use

```
julia --project=. p1.jl M_sys_125x125.txt V_sys_125x1.txt
```

This will print the number of iterations needed for convergence, and the solution vector.

Exercise 2 Application to image processing — Feature 2

1. **Implementation of basic operators**: In a `Python` script, write the following functions:
   - `forward_derivative_x`: which implements the forward differences with respect to the first variable of a 2D function.
   - `forward_derivative_y`: which implements the forward differences with respect to the second variable of a 2D function.
   - `backward_derivative_x`: which implements the backward differences with respect to the first variable of a 2D function.
   - `backward_derivative_y`: which implements the backward differences with respect to the second variable of a 2D function.

Set the values of the function to zero where they are not well-defined (it occurs at the boundary of the domain).

From these operators define:

- a function `gradient` which constructs the gradient of a 2D function using forward derivatives.
- a function `divergence` which constructs the divergence operator of a 2D vector field using backward derivatives.

2. **Image denoising**. Let u_0 be a grey-level image of size $M \times N$. A standard model to remove the noise in u_0 consists of minimizing the function

$$E(u) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \frac{1}{2}(u(x,y) - u_0(x,y))^2 + \gamma \sqrt{\|\nabla u(x,y)\|^2 + \varepsilon} \tag{2}$$

where $\gamma, \varepsilon > 0$.

The gradient of the function E at u is

$$\nabla E(u) = u - u_0 + \gamma \operatorname{div}\left(\frac{\nabla u}{\sqrt{\|\nabla u\|^2 + \varepsilon}}\right) \tag{3}$$

Task: Implement the gradient descent algorithm to minimize Equation 2 where:

- the image u_0 is `noisy2y_kodim23_greylevel.png`,
- the initial condition of the gradient descent is u_0,
- the step size α of the gradient descent is 0.0001,
- the number of iterations is 15000,
- $\gamma = 40$,
- $\eta = 0.0001$.

To execute the program, use

```
julia --project=. p2.jl
```

This will initiate the gradient descent algorithm, and after some time, will save the output denoised image. The following are some examples.


Figure 1: Noisy image.


Figure 2: Denoised with $\gamma = 40$.


Figure 3: Denoised with $\gamma = 10$.


Figure 4: Denoised with $\gamma = 0.1$.

We observe that γ correctly applies a greater blur to the image when its magnitude increases.

Exercise 3 Newton's method Feature 3

Let $f \in C^2(\mathbb{R}^n)$, and $g = f \circ A$, where $A \in \mathbb{R}^{n \times n}$ is invertible. Let us consider the sequence $\{x_k\}$ generated by Newton's method to minimize f and $\{y_k\}$ the sequence generated by Newton's method to minimize g, and where $x_0 = Ay_0$.

1. Establish a relation between the sequences $\{x_k\}$ and $\{y_k\}$.
2. Conclude.

1. Newton's method generates a sequence for minimizing $f(x)$ given by

$$x_{k+1} = x_k - \left[\nabla^2 f(x_k)\right]^{-1} \nabla f(x_k) \tag{4}$$

The sequence for minimizing $g(y)$ is given by

$$y_{k+1} = y_k - \left[\nabla^2 g(y_k)\right]^{-1} \nabla g(y_k) \tag{5}$$

By the Chain Rule, one has

$$\nabla g(y) = A^\top \nabla f(Ay) \tag{6}$$

and

$$\nabla^2 g(y) = A^\top \nabla^2 f(Ay) A. \tag{7}$$

Therefore, we have

$$\begin{aligned} y_{k+1} &= y_k - \left[A^\top \nabla^2 f(Ay_k) A\right]^{-1} A^\top \nabla f(Ay_k) \\ &= y_k - A^{-1} \nabla^2 f(Ay_k)^{-1} \left(A^\top\right)^{-1} A^\top \nabla f(Ay_k) \\ &= y_k - A^{-1} \nabla^2 f(Ay_k)^{-1} \nabla f(Ay_k) \end{aligned} \tag{8}$$

Multiplying by A on both sides yields

$$Ay_{k+1} = Ay_k - \nabla^2 f(Ay_k)^{-1} \nabla f(Ay_k) \tag{9}$$

We will prove by induction that $x_k = Ay_k$. The base case is given ($x_0 = Ay_0$), thus we assume it to be true for k. Then, one has, by substituting $x_k = Ay_k$ into the formula for $k+1$,

$$Ay_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k) = x_{k+1}. \tag{10}$$

This proves the induction hypothesis.

2. We conclude that Newton's method is affine invariant. The sequence of points remains physicially identical to the original space.

Exercise 4 Coercivity — Feature 4

1. Are the following functions coercive? Justify your answer.

$$f_1(x_1, x_2) = x_1^2 - 2x_1x_2 + x_2^2, \quad f_2(x_1, x_2) = x_1^4 + x_2^4 - 3x_1x_2 \tag{11}$$

2. Let us consider the function $f : \mathbb{R}^n \to \mathbb{R}$ given by

$$f(x) = \|Jx - y\|^2 + \alpha\|x\|^2, \quad \alpha \in \mathbb{R}, \tag{12}$$

where $J \in \mathbb{R}^{n \times n}$ and $y \in \mathbb{R}^n$. Under which conditions on J and α is f coercive?

1. To see that f_1 is **not** coercive, we take $x_1 = x_2$, and note that in this direction,

$$\lim_{|x_1| \to \infty} f_1(x_1, x_1) = \lim_{|x_1| \to \infty} 0 = 0 \neq \infty. \tag{13}$$

To prove that f_2 is coercive, we show that the lower bound of f_2 approaches infinity when the norm of the input approaches infinity. First, we see that

$$\begin{aligned} \left(x_1^2 - x_2^2\right)^2 \geq 0 &\Leftrightarrow x_1^4 + x_2^4 \geq 2x_1^2x_2^2 \\ &\Leftrightarrow 2\left(x_1^4 + x_2^4\right) \geq \left(x_1^2 + x_2^2\right)^2 = \|x\|^4 \end{aligned} \tag{14}$$

Now, by the AM-GM inequality,

$$|x_1x_2| \leq \frac{1}{2}\left(x_1^2 + x_2^2\right) = \frac{1}{2}\|x\|^2. \tag{15}$$

Therefore,

$$-3x_1x_2 \geq -3|x_1x_2| \geq -\frac{3}{2}(x_1^2 + x_2^2) = -\frac{3}{2}\|x\|^2 \tag{16}$$

We may then bound f_2:

$$f_2(x_1, x_2) \geq \frac{1}{2}\|x\|^4 - \frac{3}{2}\|x\|^2 \tag{17}$$

Taking the limit as $\|x\| \to \infty$, the 4th order term dominates the 2nd degree term. Therefore, at infinity, f_2 is bounded below by ∞, proving the assertion.

2. We expand the expression:

$$\begin{aligned} f(x) &= (Jx - y)^\top (Jx - y) + \alpha x^\top x \\ &= x^\top J^\top Jx - 2y^\top Jx + y^\top y + \alpha x^\top x \\ &= x^\top (J^\top J + \alpha I)x - 2y^\top Jx + \|y\|^2 \end{aligned} \tag{18}$$

By Variant 5, we must have $J^\top J + \alpha I \succ 0$. Therefore, if $\alpha > 0$, any J guarantees coerciveness. For $\alpha = 0$, we need

$$\lambda_{\min}(J^\top J) > 0, \tag{19}$$

and for $\alpha < 0$, we need

$$\lambda_{\min}(J^\top J) > \alpha, \tag{20}$$

and $\ker(J) = \{0\}$.

Lemma Variant 5

The function $f(x) = x^\top Ax + bx + c$ is coercive if and only if $A \succ 0$.

Proof. ($\leq$) Let $\lambda_{\min}$ be the minimum eigenvalue of A. By hypothesis, $A \succ 0$, so $\lambda_{\min} > 0$. Using Rayleigh's quotient for symmetric matrices, we have a lower bound

$$x^\top Ax \geq \lambda_{\min}\|x\|^2. \tag{21}$$

By Cauchy-Schwartz, one has

$$b^\top x \geq -\|b\|\|x\|. \tag{22}$$

Therefore,

$$f(x) \geq \lambda_{\min}\|x\|^2 - \|b\|\|x\| + c. \tag{23}$$

Since $\lambda_{\min} > 0$ and the dominance of the quadratic term, the function is bounded below by ∞ when $\|x\| \to \infty$.

($\Rightarrow$) By contrapositive, assume A is not strictly positive definite, then by symmetry of A, its eigenvalues are real. Therefore, it must be that at least one $\lambda \leq 0$. Let $v \in S^n$ be the associated eigenvector, then for $x = tv$, with $t \in \mathbb{R}_+$ one has

$$\begin{aligned}\lim_{t\to\infty} f(tv) &= \lim_{t\to\infty} t^2 v^\top Av + tb^\top v + c \\ &= \lim_{t\to\infty} t^2\lambda\|v\|^2 + tb^\top v + c \\ &= \lim_{t\to\infty} t^2\lambda + tb^\top v + c.\end{aligned} \tag{24}$$

In the case $\lambda < 0$, the limit goes to $-\infty$. If $\lambda = 0$, then either $b^\top v \leq 0$ or $b^\top v > 0$. In the former, the limit goes to $-\infty$. In the latter, we change t with $-t$, so we end up with the previous case. This proves the statement.

$\square$